
System 800xA Safety

Select I/O

Reliability and Availability





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About This Book

General

This book contains information necessary to support dependability studies and availability and reliability calculations related to the use of Select I/O in Safety Instrumented Systems (SIS).

A brief introduction to ABB reliability data methodology is given together with an overview of definitions and nomenclature.

The main parts of the book addresses safety reliability.

Document Conventions

Microsoft Windows conventions are normally used for the standard presentation of material when entering text, key sequences, prompts, messages, menu items, screen elements, etc.

Warning, Caution, Information, and Tip Icons

This publication includes **Warning**, **Caution**, and **Information** where appropriate to point out safety related or other important information. It also includes **Tip** to point out useful hints to the reader. The corresponding symbols should be interpreted as follows:



Electrical warning icon indicates the presence of a hazard which could result in *electrical shock*.



Warning icon indicates the presence of a hazard which could result in *personal injury*.



Caution icon indicates important information or warning related to the concept discussed in the text. It might indicate the presence of a hazard which could result in *corruption of software or damage to equipment/property*.



Information icon alerts the reader to pertinent facts and conditions.



Tip icon indicates advice on, for example, how to design your project or how to use a certain function

Although **Warning** hazards are related to personal injury, and **Caution** hazards are associated with equipment or property damage, it should be understood that operation of damaged equipment could, under certain operational conditions, result in degraded process performance leading to personal injury or death. Therefore, **fully comply** with all **Warning** and **Caution** notices.

Terminology

A complete and comprehensive list of terms is included in *System 800xA Terminology and Acronyms (3BSE089190)*. The listing includes terms and definitions that apply to the 800xA System where the usage is different from commonly accepted industry standard definitions.

Term/Acronym	Description
AC 800M HI	AC 800M High Integrity
BPCS	Basic Process Control System
Channel	Element or group of elements that independently perform(s) a function
Common cause failure	Failure which is the result of one or more events, causing coincident failures of two or more separate channels in a multiple channel system, leading to system failure
DC	Diagnostic Coverage
FCI	Field Communication Interface
FMEDA	Failure Mode, Effect and Diagnostic Analysis

Term/Acronym	Description
FTB	Field Termination Block
HFT	Hardware Fault Tolerance
Logic Solver	That portion of either a BPCS or SIS that performs one or more logic function(s). Logic Solver consists of Input Board, PU and Out Board. Note. Sensors and final elements are not part of the logic solver.
PFD	Probability of Dangerous Failure on Demand
PFH	Average Frequency of Dangerous Failure per Hour
Proof Test	Test performed to reveal undetected faults in a safety instrumented system so that, if necessary, the system can be restored to its designed functionality
PST	Process Safety Time
PTB	Power Injection Terminal Block
PU	Processor Unit. In this document PU refers to assembly of Processor and Supervisory Modules. This is a part of the logic solver as defined in IEC61508.
SFF	Safe Failure Fraction
SIF	Safety Instrumented Function
SIS	Safety Instrumented System
SD	Bulk power supply (main power supply)
TU	Connection unit
1oo2D	1 out of 2 Channel Architecture with Diagnostics

Related Documentation

Whenever a reference to a specific instruction is made, the instruction number is included in the reference.

Category	Title	Description
Hardware	3BSE034876	AC 800M High Integrity Reliability and Availability
	3BSE088970	Select I/O - Safety Manual
	3BSE082727	Select I/O - Getting Started and Installation
	3BSE082728	Select I/O - Modules and Termination Units
Formulas	IEC61508 part 2 and 6 Second Edition 2010	Functional Safety, international standard
	IEC/TR 62380 First Edition	Reliability data handbook Universal model for reliability prediction of electronics components, PCBs and equipment

1 Introduction

1.1 Introduction

This document includes methods and data to be used in the calculations of system reliability both from an availability as well as from a safety reliability viewpoint. Dependencies from a safety reliability viewpoint are illustrated in reliability block diagrams.

1.2 Calculation Methods

The dependability calculation methods used within ABB for the calculation of reliability complies with IEC/TR 62380 and Part 6 of IEC61508.

Quite conservative predictions can be expected from the very beginning of product life cycle. Field observations of failure rates in typical industrial applications will influence the module data given and make the data compatible to real life experiences. Improvements of the quality of the data will then be achieved gradually.

The failure rates for all circuit boards and units are calculated by a computer program. Parts count method is used. When needed, considerations are taken to actual loads. Certain components, measurement points etc. not essential for continuous operation are not included in the calculations.

The reliability data for a circuit board or a system component is stated outgoing from statistical calculation methods.

Failure rates are calculated and documented through failure mode effect and diagnostics analysis (FMEDA) and approved by TÜV.

1.3 Defined Failure Types

In this document the failure rates are divided into different types; the total failure rate λ_{TOT} , the functional failure rate λ , the dangerous undetected failure rate λ_{DU} , the dangerous detected failure rate λ_{DD} , the safe undetected failure rate λ_{SU} and the safe detected failure rate λ_{SD} . The failure rate is noted in failures per hours (1/h) in this document unless noted.

Total failure rate λ_{TOT} : A general failure in a circuit board or a system component. No considerations is given to the consequences of the failure. The failure rate for the failure type is referred to as the "Total failure rate". It is the sum of the failure rates of virtually all of the electronic components in the unit, (parts count method). It is used for maintenance predictions (spare part calculations).

Functional failure rate λ : Failure within a system component which disables complete or important parts of a system function concerned but does not affect associated system functions. Only failures which affect functions necessary for required operation are normally included.

Example: A communication is lost due to an interface error. However, the superior system function which include the communication interface is not affected. It all depends on how vital the actual communication is for the total system function.

Safe failures (λ_S): Failure within a system component which does not affect the safety integrity of the system function concerned.

Safe Detectable failures (λ_{SD}): The portion of "Safe failures" that is detected automatically by the internal diagnostics.

Safe Undetectable failures (λ_{SU}): The portion of the "Safe failures" that is not detected automatically by the internal diagnostics.

Dangerous failures (λ_D): Failure within a system component which affect the safety integrity of the system function concerned.

Dangerous Detectable failures (λ_{DD}): The portion of "Dangerous failures" that is detected and reacted upon automatically by the internal diagnostics.

Dangerous Undetectable failures (λ_{DU}): The portion of the "Dangerous failures" that is not detected automatically by the internal diagnostics. These are the most critical failures in safety related systems.

No Effect failure ($\lambda_{\#}$): Failure of an element that plays a part in implementing the safety function but has no direct effect on the safety function.

No Part failure ($\lambda_{\cdot}$): Failure of a component that plays no part in implementing the safety function.

1.4 Environment

The failure rates are calculated for industrial environment. The requirement for industrial data is stated in *Select I/O - Getting Started and Installation, Appendix A (3BSE082727)*. The ambient temperature in these calculation is assumed to be 35°C.

For higher ambient temperatures a correction factor can be used. If needed the factor can be applied to the numbers in [Table 2](#) and [Appendix A, Component Reliability Data](#).

When applied the correction will then give reliability number for higher ambient temperatures.

For ambient temperature of 55°C the Lambda values shall be multiplied with 2.0

For ambient temperature of 70°C the Lambda values shall be multiplied with 3.0

1.5 Nomenclature

Failure rate; λ states the mean number of failures per hour.

Mean Time to Failure; $MTTF = \frac{1}{\lambda}$ [hours]

Mean Time Between Failures; $MTBF = MTTF + MTTR$ [hours] [1]

Availability; $A = \frac{MTTF}{MTTF + MTTR} \times 100 = \frac{1}{1 + \sum \lambda_x MTTR}$ [%] [2]

MTTR, Mean Time To Restoration, is the average time required to move from unsuccessful operation to successful operation. The definition includes the time required to detect that a failure has occurred, and has been detected and identified, as well as the time required to make the repair.

MTTR used in the calculation of redundant functions is 8 hours.

Safety Integrity is the average probability of a safety instrumented system satisfactorily performing the required safety instrumented functions under all the stated conditions within a stated period of time.

Safety Integrity Level (SIL) is the discrete level (one out of four) for specifying the safety integrity requirements of the safety instrumented functions to be allocated to the safety instrumented systems. SIL 4 has the highest level of safety integrity; SIL 1 has the lowest.

Safe Failure Fraction (SFF) is the fraction of the overall random hardware failure rate of a device that results in either a safe failure or a dangerous detected failure.

$$SFF = \frac{\lambda_S + \lambda_{DD}}{\lambda_D + \lambda_S} \quad [4]$$

Safe State: State of the equipment under control when safety is achieved.

Diagnostic Coverage (DC)[%] and Safe Failure Fraction (SFF)[%] is calculated according to the methods given in Part 2 and the user guidelines given in Part 6 of the IEC 61508.

$$DC = \frac{\lambda_{DD}}{\lambda_D} \quad [5]$$

Common cause failure (CCF) is a failure, which is the result of one or more events, causing failures of two or more separate channels in a multiple channel system, leading to system failure.

Quantification of hardware-related common cause failures, expressed as a β -factor, is described in Annex D in Part 6 of IEC 61508. A β -factor of 1% is used in relevant calculations in this document.

Hardware Fault Tolerance is the ability of a functional unit to continue to perform a required function in the presence of faults or errors. Hardware fault tolerance with respect to dangerous failures is the ability of a functional unit to achieve safe state in the presence of a dangerous failure or error.

Proof Tests (T) are required to reveal hidden dangerous failures. The interval between such tests is of a certain importance when calculating safety reliability (PFD). The proof test interval for the logic solver is 20 years, that value is used in this document, unless specifically noted. A shorter Proof test interval can be used but the the calculations will have to be remade. A Proof test interval of 20 year is the upper limit.

Probability of Dangerous Failure on Demand (PFD) and **Probability of Dangerous Failure per Hour** (PFH) is used for reliability calculations that reflect potential effect from system failures on functional safety.

Probability of Dangerous Failure on Demand (PFD) is calculated according to the formula for an architecture in low demand mode of operation as defined in Annex B of Part 6 in the IEC 61508.



The built-in diagnostics in the AC 800M HI controller ensures that the portion of dangerous failures that is detected by the diagnostics will not prevent any safety function within the logic solver from maintaining or achieving safe state. The formula in IEC61508-6 Annex B, for 1oo1. As the system reacts on detected failures within FDRT by bringing the EUC to the safe state the time after fault detection and reaction can not contribute to the PFD value. Therefore the formula in IEC61508-6 Annex B is modified accordingly [6]. This will result in a more conservative value compared to the formula in the standard

$$PFD = \lambda_{DU} \times T \quad [6]$$

PFD is used for functions in “Low demand mode of operation”.

Probability of dangerous Failure per Hour (PFH) is calculated according to formulas for an architecture in high demand/continuous mode of operation as defined in Annex B of Part 6 of the IEC 61508.

$$PFH = \lambda_{DU} \quad [7]$$

1.6 AC 800M HI Safety Integrity

1.6.1 General

Select I/O has a basic system design that complies with IEC61508 in a 1oo2D architecture for SIL3. Individual system modules are designed in compliance with IEC61508 SIL 3 hardware integrity requirements as defined in [Table 1](#) below.

Select I/O is certified for use in up to SIL 3 applications, see [Table 1](#) and includes data from *AC 800M High Integrity Reliability and Availability manual (3BSE034876)* see table note ⁽¹⁾ for reference however for completeness that manual have to be consulted.

Table 1. Select I/O Safety Integrity

Sub-unit	SIL	HW Fault Tolerance
Generic I/O Module GIS880/TUS810	SIL 3	1
Analog Input Module AIS880	SIL 3	1
Analog Output Module AOS880	SIL 3	1
Digital Input Module DIS880	SIL 3	1
Digital Output Module DOS880	SIL 3	1
Digital Output Module DOS885	SIL 3	1

Table 1. Select I/O Safety Integrity

Sub-unit	SIL	HW Fault Tolerance
AC 800M HI PU ⁽¹⁾ - PM867/TP830 - SM812/TP868	SIL 2	0
AC 800M HI PU ⁽¹⁾ - PM867/TP830 - SM812/TP868	SIL 3	1

(1) AC 800M High Integrity Reliability and Availability
(3BSE034876)

1.6.2 BPCS Risk Reduction with AC 800M HI

Select I/O and AC 800M HI is certified for use in safety applications requiring up to SIL3 compliance. AC 800M HI application software for Safety Instrumented Functions shall be allocated and implemented in SIL marked Applications (*Control Builder M SIL applications*).

1.6.3 Safety Reliability Block Diagrams

Select I/O and AC 800M HI SIL3 1oo2D Single Controller

Reliability block diagram for the SIL3 AC 800M HI 1oo2D controller is shown below. The SIL3 architecture is different from the SIL2 because the application is now executing in both the PM and SM.

Like in the formulas 8, 9 and 10 of the SIL2 system that can be found in *AC 800M High Integrity Reliability and Availability (3BSE034876)*, the part that models the redundancy of the application execution can be reduced to the part that models the Common Cause Failure.

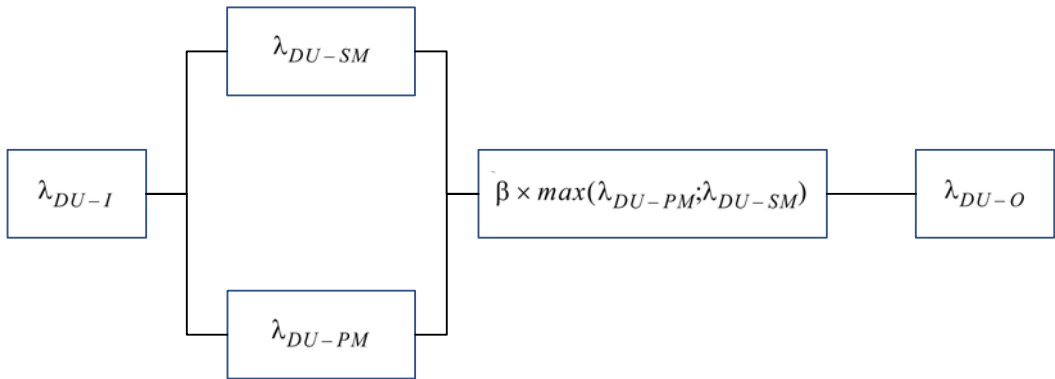


Figure 1. Reliability block diagram, SIL3 AC 800M HI 1oo2D Single Controller

Therefore the formulas are as shown in equations [11] and [12].

$$PFD = \lambda_{DU-I} \times T + \beta \times \max(\lambda_{DU-PM}; \lambda_{DU-SM}) \times T + \lambda_{DU-O} \times T \quad [11]$$

$$PFH = \lambda_{DU-I} + \beta \times \max(\lambda_{DU-PM}; \lambda_{DU-SM}) + \lambda_{DU-O} \quad [12]$$

1.6.4 Safety Reliability Calculations

The total PFD or PFH figures for a safety function can be calculated as the sum of the individual sub-system component PFD or PFH figures.

$$\text{PFD} = \text{PFD}_1 + \text{PFD}_2 + \text{PFD}_3 + \dots + \text{PFD}_i$$

$$\text{PFH} = \text{PFH}_1 + \text{PFH}_2 + \text{PFH}_3 + \dots + \text{PFH}_i$$

1.6.5 Calculations with Respect to Maintenance Effort

For calculation of maintenance effort, the total failure rate of the individual modules is used.

1.7 AC 800M HI Typical Configurations

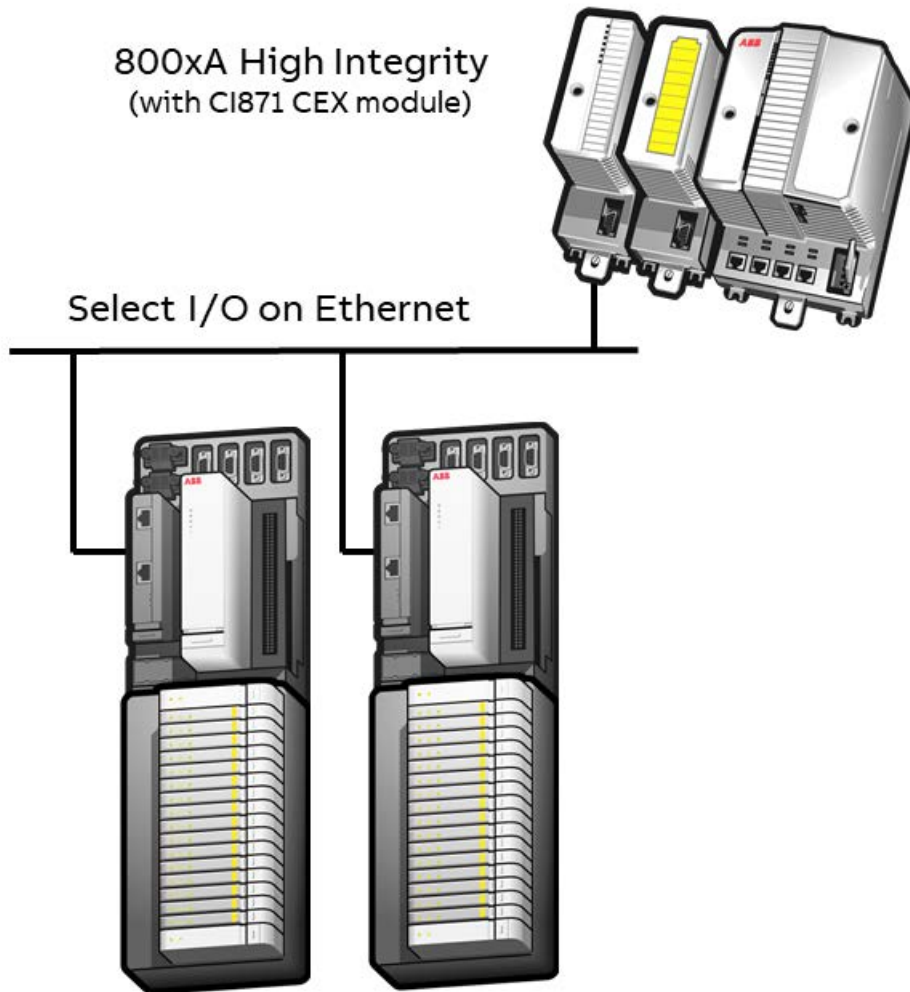


Figure 2. Select I/O System Configuration

2 Safety Reliability Calculations

2.1 General

The calculation methodology for probability of dangerous failure on demand (PFD) and probability of dangerous failure per hour (PFH) is based on formulas in IEC 61508.

Select I/O SIL3 has a 1oo2D system architecture.

Select I/O is communicating with AC 800M I/O HI via communication system that is certified for SIL3 applications. A safety layer is made on top of platform specific protocols and communication devices, thus excluding safety critical aspects of such components from the safety reliability calculations.

The Select I/O has been designed for easy adoption and scalability with respect to availability.

2.2 Configuration

Select I/O is certified for up to SIL3 applications in single configuration comprising of single AC 800M HI PU, communication and single I/O modules.

Redundancy can be implemented individually according to user documentation on a modular basis without influencing the safety integrity:

- AC 800M HI PU:
 - PM867 Processor Module
 - BC810 CEX bus interconnection Module
 - SM812 Supervisory Module
 - CI871
- CI845, TC810 and TU865
- HI880
- GIS880, TUS810
- AIS880, AOS880, DIS880, DOS880 and DOS885 I/O modules
- Power supply¹

1. These components do not contribute to dangerous undetected failures at system level because only SELV power supplies are allowed according to *Select I/O - Safety Manual (3BSE088970)*.

2.3 AC 800M HI Safety Reliability Data

For comparison of products from different manufacturers it is important to use comparable data.

In projects with multidiscipline engineering it is often necessary to distribute a PFD budget for the different subsystems. The following distribution is widely used:

- Initiator subsystem: 35%
- Logic Solver subsystem: 15%
- Final Element subsystem: 50%

In [Table 2](#) the reliability data for the AC800M HI PU are included for completeness but *AC 800M High Integrity Reliability and Availability (3BSE034876)* must always be consulted when calculating data for a complete system.

Table 2. AC 800M HI Safety reliability data

Variant	Including	λ_{du}	PFD	PFH
PM867 Single PU (SIL2) ⁽¹⁾ Processor Module Termination Plate Supervisory module Termination Plate	1 x PM867 1 x TP830 1 x SM812 1 x TP868	1.55E-08	2.83E-5 SIL2	1.61E-10 SIL2
PM867 Single PU (SIL3) ⁽¹⁾ Processor Module Termination Plate Supervisory Module Termination Plate	1 x PM867 1 x TP830 1 x SM812 1 x TP868	1.55E-08	1.41E-5 SIL3	8.07E-11 SIL3
Select I/O ⁽²⁾ Generic I/O module GIS880 Module Termination Unit (MTU) ⁽³⁾ Power Injection Terminal Block	GIS880 TUS810 2 x PTB810	3.39E-11	5.94E-06	3.39E-11
CI871 PROFINET ⁽¹⁾ Communication Board Termination Plate	1 x CI871 1 x TP867	4.93E-12	8.64E-07	4.93E-12
Analog Input channel AIS880 Field Terminal Block	AIS880 FTB810	4.05E-11	7.10E-06	4.05E-11
Analog Output channel AOS880 Field Terminal Block	AOS880 FTB810	4.60E-11	8.06E-06	4.60E-11
Digital Input channel DIS880 Field Terminal Block	DIS880 FTB810	2.52E-11	4.41E-06	2.52E-11
Digital Output channel DOS880 Field Terminal Block	DOS880 FTB810	3.51E-11	6.15E-06	3.51E-11

Table 2. AC 800M HI Safety reliability data

Variant	Including	λ_{du}	PFD	PFH
Digital Output channel DOS885 Field Terminal Block	DOS885 FTB810	3.51E-11	6.15E-06	3.51E-11
High Integrity module	HI880	0.00E+00	0.00E+00	0.00E+00
Field Communication Interface (FCI) Ethernet Adapter Module Termination Unit (MTU ⁽³⁾)	CI845 TC810 TU865	1.76E-11	3.08E-06	1.76E-11
Power supply ⁽⁴⁾	SD8xx	0.00E+00	0.00E+00	0.00E+00

- (1) Details can be found in *AC 800M High Integrity Reliability and Availability (3BSE034876)*
- (2) Each TUS810 can hold up to 16 SCM:s in any combination.
- (3) The calculations in [Table 2](#) uses the data for MTU:s from [Table 8](#) in [Appendix A, Component Reliability Data](#).
- (4) The SD8xx is included for completeness but do not have any dangerous failures.

2.4 Calculating Probability of Dangerous Failure on Demand (PFD)

Probability of dangerous failure on demand calculations are made with respect to individual safety functions in low demand systems to have a quantitative measure on safety integrity. The calculation shall include the whole control function, initiator subsystem (sensors), logic solver subsystem (processor and I/O) and the final element subsystem (actuators). *AC 800M High Integrity Reliability and Availability (3BSE034876)* have to be consulted for the AC 800M HI controller.

IEC 61508 states the range and maximum figures for probability of dangerous failure on demand (PFD_{TOTAL}) in safety instrumented functions for different Safety Integrity Levels (SIL). See [Table 3](#) below.

Table 3. Target failure measures for safety instrumented functions in low demand mode of operation

IEC61508 requirements		Initiator subsystem	Logic Solver subsystem	Final Element subsystem
SIL	PFD	35%	15%	50%
4	$10^{-4} - 10^{-5}$	3.5E-5 - 3.5E-6	1.5E-5 - 1.5E-6	5.0E-5 - 5.0E-6
3	$10^{-3} - 10^{-4}$	3.5E-4 - 3.5E-5	1.5E-4 - 1.5E-5	5.0E-4 - 5.0E-5
2	$10^{-2} - 10^{-3}$	3.5E-3 - 3.5E-4	1.5E-3 - 1.5E-4	5.0E-3 - 5.0E-4
1	$10^{-1} - 10^{-2}$	3.5E-2 - 3.5E-3	1.5E-2 - 1.5E-3	5.0E-2 - 5.0E-3

2 Safety Reliability Calculations

2.4 Calculating Probability of Dangerous Failure on Demand (PFD)

PFD figures for the most common AC 800M HI logic solver loop including Select I/O configurations are shown in [Table 4](#) below.

Table 4. AC 800M HI Logic Solver safety integrity in low demand mode of operation

SIF Architecture	SIL failure measure	
	IEC/TR 62380	
	SIL2	SIL3
	PFD	PFD
1DIS-PU-1DOS	4.87E-05	3.45E-05
1AIS-PU-1DOS	5.14E-05	3.72E-05
1DIS-PU-1AOS	5.06E-05	3.64E-05
1AIS-PU-1AOS	5.33E-05	3.91E-05

Note: AIS, DIS = Analog/Digital inputs
PU = Processing Unit (PM and SM)
DOS, AOS= Digital/Analog outputs

2.5 Calculating Probability of Dangerous Failure per Hour (PFH)

In the process industry very few safety functions are normally classified to be in high demand or continuous mode of operation. In the manufacturing industry it tends to be a higher number of applications with safety functions in high demand or continuous mode of operation.

Target failure measure for safety integrity of safety functions operating in high demand or continuous mode of operation is defined as the probability of dangerous failures per hour (PFH).

IEC 61508 states the range and maximum figures for probability of dangerous failure per hour (PFH) in safety instrumented functions for different Safety Integrity Levels (SIL). See [Table 5](#) below.

Table 5. Target failure measures for safety instrumented functions in high demand or continuous mode of operation

SIL	PFH	Initiator subsystem	Logic Solver subsystem	Final Element subsystem
		35%	15%	50%
4	$10^{-8} - 10^{-9}$	3.5E-9 - 3.5E-10	1.5E-9 - 1.5E-10	5.0E-9 - 5.0E-10
3	$10^{-7} - 10^{-8}$	3.5E-8 - 3.5E-9	1.5E-8 - 1.5E-9	5.0E-8 - 5.0E-9
2	$10^{-6} - 10^{-7}$	3.5E-7 - 3.5E-8	1.5E-7 - 1.5E-8	5.0E-7 - 5.0E-8
1	$10^{-5} - 10^{-6}$	3.5E-6 - 3.5E-7	1.5E-6 - 1.5E-7	5.0E-6 - 5.0E-7

2 Safety Reliability Calculations

2.5 Calculating Probability of Dangerous Failure per Hour (PFH)

PFH figures for the most common AC 800M HI logic solver loop including Select I/O configurations are shown in [Table 6](#) below.

Table 6. AC 800M HI Logic Solver safety integrity in high demand mode of operation

SIF Architecture	SIL reliability data	
	IEC/TR 62380	
	SIL2	SIL3
	PFH	PFH
1DIS-PU-1DOS	2.78E-10	1.97E-10
1AIS-PU-1DOS	2.93E-10	2.13E-10
1DIS-PU-1AOS	2.89E-10	2.08E-10
1AIS-PU-1AOS	3.04E-10	2.24E-10

Note: AIS, DIS = Analog/Digital inputs
PU = Processing Unit (PM and SM)
DOS, AOS= Digital/Analog outputs

2 Safety Reliability Calculations

2.5 Calculating Probability of Dangerous Failure per Hour (PFH)

Appendix A Component Reliability Data

The following [Table 7](#) and [Table 8](#) give detailed figures for safety certified and safety relevant HW components. It is not always possible to use the numbers found in this appendix to calculate the values in the System Availability calculation section. The calculated numbers are based on additional documented assumptions in each configuration..

Table 7. Safety certified Select I/O HW component reliability data.

Module	Description	I_{sd}	I_{su}	I_{dd}	I_{du}	$I_{\#}$	$I_{\cdot}$
GIS880	Generic I/O Module	6.01E-07	1.08E-07	3.56E-07	3.36E-11	2.15E-07	0.00E+00
AIS880	Analog Input Module	1.24E-07	3.82E-08	6.58E-07	4.00E-11	3.12E-07	7.61E-08
AOS880	Analog Output Module	1.84E-07	3.46E-08	8.53E-07	4.55E-11	3.14E-07	0.00E+00
DIS880	Digital Input Module	1.84E-07	6.02E-08	5.13E-07	2.46E-11	2.62E-07	0.00E+00
DOS880	Digital Output Module	4.15E-07	6.95E-08	5.31E-07	3.46E-11	2.62E-07	0.00E+00
DOS885	Digital Output Module	4.15E-07	6.95E-08	5.31E-07	3.46E-11	2.62E-07	0.00E+00
HI880	High Integrity Module	1.07E-07	8.93E-12	0.00E+00	0.00E+00	5.37E-08	0.00E+00

Table 8. Safety relevant HW component reliability data.

Module	Description	λ_{sd}	λ_{su}	λ_{dd}	λ_{du}	$\lambda_{\#}$	λ_{-}
CI845	Fieldbus Communication Interface Module	1.51E-06	6.00E-08	2.74E-08	2.74E-12	5.55E-07	0.00E+00
TC810	Ethernet Adapter	3.20E-07	5.51E-08	4.56E-09	4.56E-13	1.17E-07	2.32E-08
TU865	Termination Unit	9.74E-08	5.72E-08	1.44E-07	1.44E-11	5.67E-08	0.00E+00
TUS810 2xPTB810	MTU with Power Injection Terminal Block	5.87E-07	5.87E-11	8.16E-09	2.72E-13	1.52E-07	3.40E-09
FTB810	Field Terminal Block	0.00E+00	0.00E+00	5.44E-09	5.44E-13	1.36E-09	0.00E+00

Revision History

The following table lists the revision history of this User Manual.

Revision Index	Description	Date
A	First published for I/O System Version 1.0	January 2020



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